

IPL DATA ANALYSIS

Python & Power BI Based Analytics Project

Major Project – I

Data Analytics Master Program – Batch 2

Student	Yasaswini Adimulam
Project	IPL Data Analysis
Python	Pandas, NumPy, Matplotlib, Seaborn
BI Tool	Microsoft Power BI

Submitted as part of the Major Project requirements.

1. Abstract

This project analyzes Indian Premier League (IPL) match and ball-by-ball data to identify useful patterns in team performance, player performance, toss outcomes, season trends, and venue usage. Python was used for data exploration, analysis, and visualization, while Microsoft Power BI was used to create an interactive analytics dashboard. The project demonstrates a complete data analytics workflow from cleaned datasets to visual insights and business-oriented conclusions.

2. Introduction

The IPL produces a large amount of structured sports data across seasons, matches, players, teams, venues, toss decisions, runs, and wickets. Analyzing this data helps understand historical performance and provides a clear way to communicate trends through dashboards. This project combines

Python-based statistical analysis with Power BI reporting to produce a practical sports analytics solution.

3. Project Objectives

- Analyze IPL match and ball-by-ball datasets.
- Explore team, player, toss, season, and venue performance.
- Create meaningful Python visualizations from the analyzed data.
- Develop an interactive Power BI dashboard with KPIs and analytical visuals.
- Generate clear business insights from IPL data.

4. Dataset Overview

The project uses cleaned IPL datasets. The match-level dataset contains 1,095 matches and 22 columns. The delivery-level dataset contains 260,920 ball-by-ball records and 17 columns. A season summary dataset contains 17 season-level records.

5. Technologies Used

Python, Pandas, NumPy, Matplotlib, Seaborn, Microsoft Power BI, and DAX were used for the project. Python handled data preparation, aggregation, analysis, and charts. Power BI handled the interactive dashboard, KPIs, filtering, and reporting.

6. Methodology

Step 1 – Data Preparation: cleaned IPL match and delivery datasets were used as the analytical source.

Step 2 – Data Exploration: dataset dimensions, columns, data types, missing values and duplicate records were inspected.

Step 3 – Analysis: match, team, player, toss, season and venue-level aggregations were calculated.

Step 4 – Visualization: Python charts were created to communicate major analytical results.

Step 5 – Power BI: match and delivery data were connected using the match identifier, DAX measures were created, and an interactive dashboard was developed.

7. Python Analysis

The Python notebook IPL_Data_Analysis.ipynb contains the complete workflow: dataset loading, exploration, cleaning checks, match analysis, player performance analysis, team analysis, toss analysis, season analysis, visualizations, and business insights.

8. Python Visualizations

Fourteen visualizations were created: Matches Played by Season; Matches Won by Team; Top 10 Run

Scorers; Top 10 Wicket Takers; Player of the Match Awards; Toss Decision Distribution; Matches Won

After Winning Toss; Season-wise Total Runs; Season-wise Total Wickets; Runs Distribution; Team Performance Comparison; Top 10 Venues by Match Count; Team Runs vs Wickets; and Correlation Heatmap.

9. Power BI Dashboard

The Power BI solution contains KPI cards and analytical visuals covering batting, bowling, team performance, season trends, toss analysis, and match results. A Season slicer was added to allow interactive filtering of the dashboard.

Key Dashboard KPIs

KPI	Project Result
Total Matches	1,095
Total Runs	Approximately 348K
Total Wickets	Approximately 13K
Total Seasons	17
Total Batsmen	673

10. DAX Measures

Total Matches = `DISTINCTCOUNT(IPL_Cleaned_Matches[id])`

Total Runs = `SUM(IPL_Cleaned_Deliveries[total_runs])`

Total Wickets = `CALCULATE(COUNTROWS(IPL_Cleaned_Deliveries),
IPL_Cleaned_Deliveries[is_wicket] = 1)`

Total Seasons = `DISTINCTCOUNT(IPL_Cleaned_Matches[season])`

11. Data Model

The Power BI model connects `IPL_Cleaned_Matches[id]` on the one side to `IPL_Cleaned_Deliveries[match_id]` on the many side. This relationship allows match-level attributes such as season to filter delivery-level calculations.

12. Key Insights

- Team wins can be compared to identify historically strong-performing teams.
- Top batsmen and bowlers can be identified using total runs and wickets.
- Toss winners also won 554 of 1,090 completed matches, approximately 50.83%, in the analyzed data.
- Season-wise runs and wickets reveal changes in scoring and wicket patterns across IPL seasons.
- Venue-level analysis identifies locations that hosted a large number of matches.
- Combining Python analysis with Power BI provides both detailed analysis and interactive reporting.

13. Conclusion

The IPL Data Analysis project successfully demonstrates a complete data analytics workflow. Cleaned IPL data was explored and analyzed using Python, fourteen analytical visualizations were created, and the results were presented through an interactive Power BI dashboard. The project provides a practical example of combining data preparation, exploratory analysis, visualization, DAX, and business insight generation.

14. Future Scope

- Add predictive models for match outcomes and player performance.
- Include advanced player form and season-over-season comparisons.
- Add more interactive Power BI drill-through pages and filters.
- Automate data refresh for newly completed IPL seasons.
- Explore machine learning techniques for win-probability prediction.

15. Final Project Files

Dataset → cleaned IPL CSV files

Python → IPL_Data_Analysis.ipynb

PowerBI → IPL_Analytics_Dashboard.pbix

Report → IPL_Project_Report.pdf

Root → README.md